E-Commerce Sales Analysis

Business Requirement

Business Scenario

An e-commerce company wants to analyze its sales performance to better understand customer purchasing behavior and improve business decision-making. The company sells products across multiple categories and regions, offering various payment methods and discounts to customers.

As a **Junior Data Analyst**, your responsibility is to analyze the e-commerce sales dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify the best-selling product categories, high-performing regions, customer preferences, and revenue trends.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide meaningful recommendations.

Project Objective

Analyze the e-commerce sales dataset to:

- Understand overall sales performance.
- Identify the highest revenue-generating product categories.
- Analyze regional sales performance.
- Study customer purchasing behavior.
- Generate business insights and recommendations to improve sales.

Dataset Description

The dataset contains information about **e-commerce sales transactions**, including order details, product categories, customer information, pricing, discounts, payment methods, delivery performance, customer ratings, and revenue.

Dataset Features

Column	Description
Order ID	Unique Order Identifier
Order Date	Date of Order
Customer ID	Unique Customer Identifier
Product Category	Category of Product Purchased
Region	Sales Region
Quantity	Quantity Purchased
Unit Price	Price per Unit
Discount	Discount Offered
Payment Method	Payment Mode Used by Customer
Delivery Days	Number of Days Taken for Delivery
Customer Rating	Customer Rating
Revenue	Total Revenue Generated

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
- Load the dataset using Pandas.
- Display the first five records.
- Display the last five records.
- Find the number of rows and columns.
- Display all column names.
- Check the data types of each column.

Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
- Check duplicate records.
- Display summary statistics.
- Find the number of unique product categories.
- Find the number of unique regions.
- Find the number of unique payment methods.

Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
- Handle missing values.
- Convert **Order Date** into Date format.
- Create a new column named **Order Year**.
- Create a new column named **Order Month**.
- Ensure **Revenue**, **Unit Price**, **Quantity**, **Discount**, **Delivery Days**, and **Customer Rating** have the correct data types.

Business Questions

Answer the following questions using Python and Pandas.

Sales Analysis

- What is the total revenue generated by the company?
- What is the average revenue per order?
- Which orders generated the highest revenue?
- Which orders generated the lowest revenue?

Product Category Analysis

- Which product category has the highest number of orders?
- Which product category generates the highest total revenue?
- Which product category has the highest average revenue per order?

Regional Analysis

- Which region has the highest number of orders?
- Which region generates the highest revenue?
- Compare revenue across different regions.

Customer Analysis

- How many unique customers have placed orders?
- Which customers have placed the highest number of orders?
- Which customers generated the highest revenue?

Payment Method Analysis

- Which payment method is used the most?
- Which payment method generates the highest revenue?
- Compare revenue across different payment methods.

Discount Analysis

- Which product categories receive the highest average discount?
- Does offering higher discounts lead to higher revenue?
- Which orders received the maximum discount?

Delivery Performance Analysis

- What is the average delivery time?
- Which region has the fastest average delivery?
- Which region has the slowest average delivery?

Customer Rating Analysis

- What is the average customer rating?
- Which product category has the highest average customer rating?
- Which region has the highest customer satisfaction?

Monthly Sales Analysis

- How many orders were placed each month?
- Which month generated the highest revenue?
- How does revenue change over time?

Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Revenue by Product Category (Bar Chart)
- Revenue by Region (Bar Chart)
- Monthly Revenue Trend (Line Chart)
- Customer Rating Distribution (Histogram)
- Payment Method Distribution (Pie Chart)

Optional Chart

- Top 10 Highest Revenue Orders (Horizontal Bar Chart)

Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which product categories contribute the most to total revenue?
- Which regions are the company's strongest markets?
- Which payment methods are preferred by customers?
- How do discounts impact sales performance?
- Which months experience the highest sales activity?

Business Recommendations

Provide **at least 3 recommendations** for the e-commerce company.

Example recommendations:

- Increase inventory and marketing for high-performing product categories.
- Focus promotional campaigns on regions with high sales potential.
- Optimize discount strategies to improve profitability.
- Improve delivery performance in regions with longer delivery times.
- Encourage customers to use the most popular and efficient payment methods.

Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
- Cleaned Dataset (if modified)
- Report
- 3 Business Recommendations
- 5-minute PowerPoint Presentation

Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	**100**